

The Digital Human Revolution

Pioneering Advanced Approaches to Hyper-Realistic Digital Characters to Drive Impact-Driven Storytelling Across Platforms and Use Cases

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ABSTRACT

This paper explores the creation of advanced 'digital humans' using 'Metahuman' technology [Fang et al. 2021] and their role in elevating storytelling across platforms. UME Studios' groundbreaking work in the virtual reality experience "Stay Alive My Son" [Pin 2000] (SAMS) [Bousis 2021] is discussed, highlighting their achievement in representing real people at high fidelity. The design and implementation process of digital humans in SAMS, along with the technical and ethical considerations [IEEE 2022] involved, are examined.

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1 INTRODUCTION

The advancement of digital human technology has revolutionized compassionate-driven storytelling in virtual reality [Olynick 2022]. This paper examines UME Studios' innovative approach to designing and implementing digital humans in Stay Alive, My Son, utilizing advanced metahuman technology to drive powerful narratives. It delves into the technical and ethical considerations, inherent in digital representations of people versus fictional characters, that shape UME's transformative approach.

2 DESIGN AND IMPLEMENTATION

Back in 2021, UME Studios faced challenges in customizing digital humans based on real individuals, as existing tools had limitations. Months of meticulous research, design, iteration, and implementation were required to humanize the digital representations of Yathay Pin and his missing son, Nawath. The technical pipeline involved modeling, remodeling, and customization using various tools such as Metahuman Creator, MeshMorpher, Z-Brush, and Blender. Attention to detail was crucial, focusing on capturing the realistic features and nuances of the characters. Advanced texturing

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Figure 1: Real Life Yathay Pin

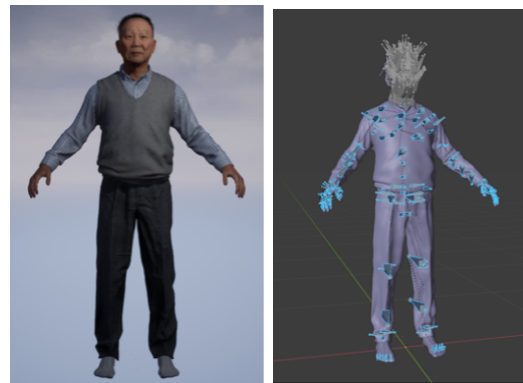


Figure 2: Metahuman Yathay Pin and Clothing (L), and Yathay's Body Rig (R).

and lighting techniques were employed to achieve hyper-realistic skin realism (See figure 1, figure 2). Particular attention was made in modeling the body, using the clothing made in Marvelous Designer to convey realism of not only the clothing Yathay wears in real life, but also muscular atrophy which occurs in the elderly to humanize the digital human. (See Figure 2).

3 CHALLENGES

Creating a believable and empathetic digital representation of a child posed specific challenges. UME Studios had to navigate anatomical proportions, facial expressions, and behavioral nuances that differed from adult characters. Ethical considerations were paramount, especially when representing a missing child. Striking a balance between narrative drive, realism, and sensitivity required



Figure 3: Real Life Nawath (L), and Metahuman Nawath (R).

careful decision-making. Because we do not know whether Nawath is still alive, Bousis felt it was inappropriate to model Nawath exactly like the real boy. To preserve the integrity of the work, elements from the actor representing Nawath were integrated, considering the uncertainty surrounding the real boy's current status. (See Figure 3).

4 CONCLUSION

The "Stay Alive My Son" [Pin 2000] project [Bousis 2021] showcases the transformative potential of advanced digital human technology in VR storytelling. The immersive experience fosters a deep connection between participants and the digital human characters, serving as a catalyst for social change. The interplay between technical innovation and ethical considerations redefines the boundaries of VR storytelling, empowering authentic narratives about real people. The integration of metahumans holds promise in promoting compassionate policy-making, enhancing human connection, and amplifying marginalized voices. By harnessing cutting-edge technology with sensitivity and care, virtual experiences and narratives can inspire empathy, drive change, and amplify unheard voices.

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